

The Potential of Learning With AI-Generated Pedagogical Agents in Instructional Videos

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ABSTRACT

With the recent advancement in technology, generative artificial intelligence (GenAI) can produce hyper-realistic multimedia content, such as audio, text, images, and videos. Although this technology has raised great concerns about its misuse and harmful applications, it holds great potential to revolutionize traditional ways of teaching and learning. The use of GenAI in education has increased markedly, however, pedagogical research on this rapidly emerging technology is yet to be studied extensively. There is an urgent need to investigate the unexamined potential of this technology. Therefore, this ongoing research will explore the potential of AI-generated pedagogical agents (PA), or avatars, in instructional videos to facilitate learning. The effects of the type of PA (AI-generated, real-life human), and voice (AI-generated, human voice) on an individual's learning outcomes, cognitive load, motivation, and attention will be studied. Findings from a pilot study provide some preliminary evidence that PA appearance influences learners' retention and cognitive load, but not attention. The type of PA influenced learners' perception of the agent's ability to facilitate learning, its human-like qualities, and its engagement level. However, it did not affect its credibility. This ongoing work will contribute to the growing understanding of the impact of AI in education, provide evidence of the efficacy of AI-generated PAs in instructional videos for learning, and narrow the gap between human-computer interaction research and education.

CCS CONCEPTS

• **Applied computing** → Education; Distance learning; Education; E-learning; Law, social and behavioral sciences; Psychology; • **Human-centered computing** → Human computer interaction (HCI); • **Computing methodologies** → Artificial intelligence.

KEYWORDS

pedagogical agents, avatars, multimedia learning, videos

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1 INTRODUCTION

Recent advancements in generative artificial intelligence (GenAI) technologies have allowed hyper-realistic, human-like content to be produced in diverse modalities, including audio, text, images, and videos [19, 29]. In the context of education, while research has found GenAI to provide opportunities for students to receive personalized learning support [8] and teachers to enhance their pedagogical efficacy [28], discussions on the applications of GenAI have remained limited to intelligent tutoring systems, chatbots, and automated grading systems [26]. Research on the pedagogical uses of this rapidly emerging technology remains minimal, and therefore there is an urgent need to investigate the unexamined potential of this technology [16]. Furthermore, a common limitation of human-computer interaction research is the need to further explore educational effectiveness [16]. And this research will address these needs.

This research aims to explore the potential of GenAI, specifically AI-generated pedagogical agents (PAs) in instructional videos, to facilitate learning. PAs are life-like characters in multimedia learning environments, usually presented on a computer screen [10]. The initial inspiration for this idea stemmed from the researcher's previous work in augmented reality (AR), which were educational experiences to assist individuals with special needs in practicing social interactions with human-like avatars. Expanding on this concept of social interaction, the researcher delves deeper into the notion of enhancing learning experiences through interactions with digital agents. This interest was particularly piqued by the social agency theory, which is "based on the idea that the learner can interpret a multimedia learning episode as either a case of information delivery or a case of social communication" [23]. Furthermore, research on PAs in multimedia learning environments has shown evidence that animated PAs in multimedia can improve learning outcomes [2] and positively impact learners' behaviors, attitudes, and motivation [3]. In recent research on synthetic virtual instructors, some evidence of its effectiveness on education was found, but researchers addressed the need to further investigate synthetic virtual instructors' effectiveness with a more sophisticated measure of learning effectiveness [16]. This has further motivated the researcher to investigate AI-generated PAs with a focus on measuring the effects of the technology on learning outcomes and cognition. In a preliminary pilot study, similar to the above-mentioned research, nearly equal learning outcomes between an AI-generated PA and a human instructor video were observed. However, one significant difference was that participants in the human instructor perceived a greater social presence of the instructor compared to participants in the AI-generated PA group. These findings have inspired the design of pilot study 1 and ongoing research, where the

social aspect is explored more extensively through an examination of appearance and voice with refined measures.

In this ongoing research, the effects of PA type (AI-generated, real-life human), and voice (AI-generated, human voice) on an individual's learning outcomes (retention, recall, and transfer), cognitive load, motivation, and attention will be examined. To the researcher's knowledge, this study will be one of the first to empirically investigate the effects of AI-generated PAs with hyper-realistic appearance and voice on learning at the same time.

2 RELATED WORK

Previous research has found that learners can interpret multimedia learning content as a case of social communication [23], and learners learn more deeply from audio words spoken in a human voice rather than a machine voice [20]. Likewise, animated PAs have also been found to improve learning outcomes [2], and positively impact learner motivations [3].

2.1 Social agency theory

The social agency theory is “based on the idea that the learner can interpret a multimedia learning episode as either a case of information delivery or a case of social communication.” [23]. According to this theory, PA's social cues (e.g. human-like gestures, image, conversation style, eye gaze, movement, and voice) in multimedia environments can replicate the human-to-human connection and are perceived similarly to a human-to-human interaction [2, 12]. This creates a feeling of social relationship between the PA and the learner, which facilitates learning by engaging deeper cognitive processes [2, 20, 22, 24].

2.2 Pedagogical agents and social cues

Previous research has found that PA social cues enhance learner social attention [23], attention [22], motivation [3], learning outcome, and cognitive load management [4, 22]. Social cues, such as voice and embodiment, in multimedia learning can “foster student motivation and meaningful learning” [14].

2.2.1 Voice. Previously, the role of PA voice has been examined by comparing human voices to machine-generated synthetic voices, and accented speech to native English speech [2, 22, 23]. Researchers investigated whether PAs with human voices contributed to a learning interest increase and better transfer, compared to PAs with machine-generated voices, and results showed that students in the human voice group outperformed the machine voice group [2]. Similarly, in another study, students performed better on a problem-solving transfer test when they learned from PAs that spoke in a human voice, compared to PAs with a machine voice [22]. Now, with GenAI producing hyper-realistic voices, there is a need to further study the effect of machine-generated voices in these contexts.

2.2.2 Embodiment. Several studies have examined the embodiment of PAs [3, 12, 25]. Students who learned about the human cardiovascular system from a narrated multimedia lesson with PAs that showed human-like gestures and pointing performed better on a transfer test, compared to those who learned from a low-embodied

PA [12]. Similarly, participants who learned about electrical circuits from a PA with human-like pointing performed better on a transfer test, compared to those who learned from a multimedia lesson with arrows [25]. Furthermore, students who learned how to use a computer software from a PA with gestures performed better on a comprehension test, compared to those who learned from PAs without gestures [3]. Embodiment is an important characteristic of PAs to promote learning, especially human-like movements, such as eye contact, facial expressions, or body gestures [14]. Now that GenAI technology can replicate human movement and facial expressions very closely to real-life humans, a further investigation of the effects of hyper-realistic embodiment affects learning outcomes is needed.

2.3 Limitations

Despite the findings of previous studies that support the idea of PAs' positive effects on learning, there are gaps in literature. Researchers have raised the need to study: human-like PAs more extensively [11], interactions between different types of PA social cues [27], motivation [11], and instructional benefits of PA in different learning contexts [30]. Another limitation identified in previous research is that most studies were conducted in traditional classroom or laboratory settings [12], which do not represent the context videos are consumed today. More specifically in the research on AI-generated synthetic virtual instructors by [16], researchers raised the need to utilize a more robust learning outcome measure in their study. To address all these gaps in literature, this research study will be one of the first to empirically investigate the effects of hyper-realistic looking (appearance) and sounding (voice) PAs affect an individual's learning outcomes, cognitive load, motivation, and attention in a non-traditional classroom or laboratory setting.

2.4 The potential of using avatars for special needs populations

Further to previous research on PAs, there is emerging evidence that supports the use of an avatar with live animation to intervene with students with autism spectrum disorder (ASD) [9]. Perhaps, this suggests we could expand the research of AI-generated PAs and connect it to assisting individuals with special needs in educational settings in a similar way. Furthermore, previous research has found ASD children tend to be visual learners [13], and more studies show the effectiveness of avatars designed to aid visual learners. Previous research has found that students with ASD are engaged by animated avatars and benefit from seeing visual models of target skills when learning social skills [6]. In another study, the use of a computer-animated avatar elicited higher motivation for students with ASD to learn, with fewer disruptive behaviors, compared to more traditional personal instruction [7]. Researchers [31] also found evidence that suggests an intervention with a humanoid avatar may change and improve the motivation to communicate via speech. It is imperative to thoroughly investigate this further, and the findings of this research will serve as a stepping stone in understanding the potential of using AI pedagogical agents to assist diverse learner needs, such as learners with special needs like ASD.

Experimental Group 1 (AI pedagogical agent + AI-generated voice)	Experimental Group 2 (AI pedagogical agent + Human voice)	Experimental Group 3 (Human instructor + AI-generated voice)	Control Group (Human instructor + Human voice)
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Figure 1: Participant groups.

3 RESEARCH APPROACH AND METHODS

The purpose of this research is to explore the potential of AI-generated PAs in instructional videos to facilitate learning, by investigating the effects of PA type (AI-generated, real-life human), and voice type (AI-generated, human voice) on learning outcomes (retention, recall, and transfer), cognitive load, motivation, and attention.

3.1 Research questions

At large, this research will address the question: Are AI-generated PAs equally capable of facilitating learning as human instructors in instructional videos?

This inquiry will be investigated by pursuing the following research objectives:

- How do AI-generated PA features influence learners' perception of agents?
- What is the effect of AI-generated PAs on learners' attention in instructional videos? How does this vary by type of PA and type of voice?
- What is the effect of AI-generated PAs on learners' retention in instructional videos? How does this vary by type of PA and type of voice?
- What is the effect of AI-generated PAs on learners' cognitive load in instructional videos? How does this vary by type of PA and type of voice?

3.2 Research design

This study will employ a 2 x 2 factorial design, with type of PA (AI-generated, human teacher), and voice (AI-generated, human voice) as the two factors to examine the comparable effects of instructional videos with AI-generated PAs vs. real-life human instructors. Participants will be randomly assigned to one of the four conditions (see Figure 1) on an online research platform, where they will watch a 3-minute video on cacao nibs (an unfamiliar topic) and answer questions afterward. The learning content will be identical, but the PA, and voice combination will vary depending on the assigned condition (see Figure 1). The key dependent measure is post-test scores (learning outcomes). Data on social connection with the PA (API), attention (eye-tracking), motivation (IMMS survey), and cognitive load (survey) will also be collected.

3.2.1 Participants. The inclusion criteria for the participant population of this study are individuals who: are adults (persons over 18 years of age), are comfortable being camera-recorded (no face, only eye movement data), and do not require assistive devices or software to use a computer.

The target sample size for the full study is 140 participants, which is 35 participants per condition (4 conditions in total). Based on a

power analysis on G*Power3.1, 128 participants is the best sample size to have an approximate power of 0.8 and effect size (Cohen's d) of 0.25 (medium effect) in a between-subject two-way ANOVA. The number of participants is slightly larger than this parameter to count for any possible attrition. Furthermore, previous studies with similar research designs had sample sizes of around 30 participants per condition, usually ranging from 20 to 34 participants per condition [2, 3, 23]. Some had even smaller sample sizes of 13 participants per condition [1].

For pilot study 1, a total of 32 adults enrolled in two higher education institutions in the state of New York participated voluntarily. However, due to attrition caused by 2 participants not completing the study, data of 30 participants were used for analysis.

3.2.2 Materials and instruments. All materials for this study will be published via an online research platform (with eye-tracking functionalities), which consists of a pre-test questionnaire, four versions of the 3-minute instructional video about cacao nibs, the cognitive load survey [17, 18], Agent Persona Instrument (API) survey [5], Instructional Materials Motivation Survey (IMSS) [15], and a learning posttest. In terms of the PAs, Figure 2 shows an example of how the human and AI-generated PAs will appear visually.

The pre-test questionnaire is a self-reported 10-item measure that will collect data on participant information on demographics, learning preference, frequency of watching educational videos, prior knowledge of the topic cacao nibs, and whether or not they have a disability. Measuring prior knowledge has been commonly conducted in previous studies [21], and is beneficial in preventing testing effects, where pre-test learning influences subsequent learning. Furthermore, the question on disability is to account for any potential observed differences based on disability and to inform future research that could be explored further. [1].

Participants will watch a 3-minute instructional video on cacao nibs (an unfamiliar topic to most people), and the learning content will be identical across all conditions, however, based on the assigned condition, participants will either watch a video with: (1) AI PA+ AI-generated voice, (2) an AI PA + human voice, (3) human instructor + AI-generated voice, or (4) human instructor + human voice. Figure 2 shows an example of how the human and AI-generated PAs will visually appear differently in the videos. While watching the video, participant eye-tracking data, such as areas of interest (AOI), fixation, and gaze points, will be collected.

The cognitive load survey contains 10 self-report items that will measure the perceived cognitive load of participants. Participants will answer 10 questions on a scale of 0 (not at all the case) to 10 (completely the case). It has been developed and used in previous studies [17, 18].

The Agent Persona Instrument (API) survey [5] contains 25 self-report items that will measure the participants' social relationship

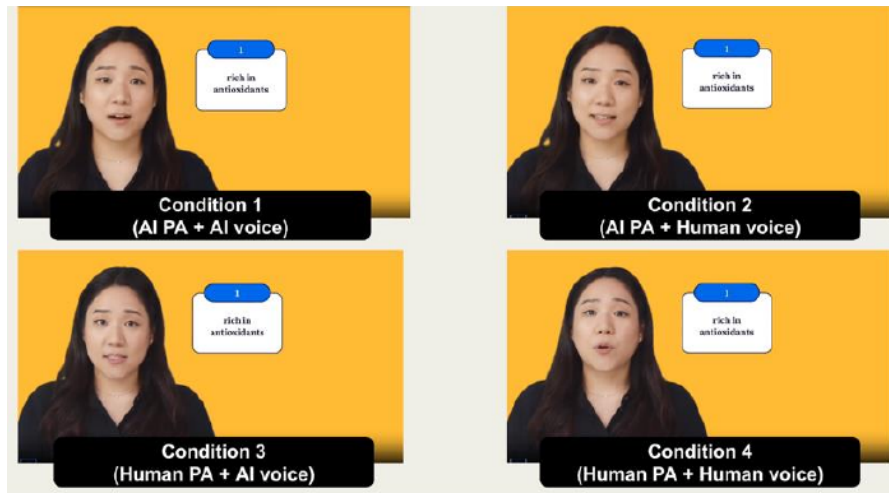


Figure 2: Video screen captures of different condition groups.

with the PA. Participants will answer questions about how well the PA facilitated learning, its credibility, how human-like it is, and how engaging it is on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

The Instructional Materials Motivation Survey (IMSS) [15] is a commonly used 36-item situational measure of learner's reactions to instructional materials based on the framework of the ARCS (Attention, Relevance, Confidence, and Satisfaction) model. Participants will answer questions about their motivation in the learning experience on a 5-point Likert scale, ranging from 1 (not true) to 5 (mostly true).

The post-test includes a retention test consisting of 5 short answer questions, and 7 true/false questions that measure the learner's recall of learning contents, followed by the transfer test comprising 2 short answer questions. Open-ended short-answer questions have been used in previous studies to measure retention and transfer [21]. Furthermore, a scoring rubric for these short answer questions was constructed by identifying key ideas from the learning content (retention test) and determining a set of acceptable answers (transfer test).

3.2.3 Procedure. Participants will follow the procedures:

1. Random assignment (4 condition groups)
2. Complete pre-test questionnaire (12 questions)
3. Watch a 3-minute stimulus (instructional video)
4. Complete the cognitive load survey (10 questions)
5. Complete Agent Persona Instrument (API; 25 questions)
6. Complete Instructional Materials Motivation Survey (IMSS; 36 questions)
7. Complete post-test survey (15 questions)

3.3 Analysis plan

Inferential statistics will be used to analyze the relationship between the key dependent measures. A two-way analysis of variance (ANOVA) will be used to examine the main effects of the

two independent variables (PA type and voice) on each dependent measure separately. In addition, the interaction effects will also be assessed whether the combined influence of PA type and voice on a dependent measure is significantly different from what would be expected based on their individual effects, on social connection, attention, motivation, cognitive load, and posttest scores. Multiple linear regression will also be used to investigate more complex relationships of the combined impact of PA type and voice on social connection, attention, motivation, cognitive load, and post-test scores.

4 RESULTS AND CONTRIBUTIONS TO DATE

As part of this ongoing research, 1 preliminary pilot study and 1 pilot study have been conducted to date.

4.1 Preliminary pilot study

In a preliminary pilot study, 2 conditions (AI PA vs. human instructor video) were compared, and nearly equal learning outcomes and cognitive load between both conditions were observed. This similarly replicated the results of research by Leiker et al. [16] on synthetic virtual instructors. However, one significant difference was that participants in the human instructor perceived a greater social presence of the instructor than participants in the AI-generated PA groups, and it has inspired the design of the currently ongoing research to examine appearance and voice more extensively.

4.2 Pilot study 1

Following the preliminary pilot study, pilot study 1 was conducted with refined measures that consisted of instruments closely related to or identical to previous research on PAs, such as Agent Persona Instrument (API), retention, cognitive load, and attention. The pilot study examined the effects of PA type (AI-generated vs. human) and voice (AI-generated vs. human) on learners' agent perception, attention, retention, and cognitive load. The mean scores

Table 1: Means and standard deviations of the dependent variables.

Dependent variable	Measures		Conditions			
			AI PA + AI Voice($n = 8$)	AI PA + human voice($n = 6$)	Human PA + AI-voice($n = 8$)	Human PA + human voice($n = 8$)
Agent Perception ^a	API	M	12.97	13.53	15.46	16.21
		SD	3.41	2.41	2.48	2.21
Attention ^b	K-coefficient	M	.32	.32	.33	.33
		SD	.18	.10	.06	.07
Retention ^c	Post-test	M	6.50	5.83	6.50	6.62
		SD	.54	1.17	.76	.52
Cognitive load ^d	Cognitive load survey	M	30.38	33.17	22.50	36.00
		SD	9.65	8.28	8.16	4.81

^a Possible range for API (0-20); ^b Range of ambient attention ($K < 0$) and focal attention ($K > 0$);

^c Possible range for retention (0-7); ^d Possible range for cognitive load (0-100).

and corresponding standard deviations of all dependent variables (agent perception, attention, retention, and cognitive load) of each condition group are presented in Table 1.

4.2.1 RQ1 - How do AI-generated PA features influence learners' perception of agents? The results of the 2 x 2 ANOVA on API showed a significant main effect of the type of PA, $F(1, 29) = 6.85, p = .015$.

A further analysis conducted on each sub-item of the agent perception measure (API), API-1 (learning facilitation) showed a significant main effect on PA, $F(1, 29) = 5.17, p = .031$, API-3 (human-likeness) showed a significant main effect on PA, $F(1, 29) = 10.80, p = .003$, and API-4 (engagement) showed a significant main effect on PA, $F(1, 29) = 6.89, p = .014$. However, no statistically significant findings were observed with API-2 (credibility).

4.2.2 RQ2 - What is the effect of AI-generated PAs on learners' attention in instructional videos? How does this vary by type of PA and type of voice? No statistically significant effects were found from the 2 x 2 ANOVA on attention.

4.2.3 RQ3 - What is the effect of AI-generated PAs on learners' retention in instructional videos? How does this vary by type of PA and type of voice? The results of the 2 x 2 ANOVA on retention showed no significant main effect on PA, $F(1, 29) = 2.05, p = .17$, no significant main effect of voice, $F(1, 29) = .96, p = .34$, and no significant interaction effect between PA and voice, $F(1, 29) = 2.05, p = .17$. However, a 2 x 2 between-groups ANCOVA on retention (with English proficiency as a covariate) indicated a significant interaction effect between PA and voice, $F(1, 29) = 4.34, p = .045$.

4.2.4 RQ4 - What is the effect of AI-generated PAs on learners' cognitive load in instructional videos? How does this vary by type of PA and type of voice? No statistically significant effects were found from the 2 x 2 ANOVA on cognitive load. However, further analysis of a 2 x 2 between-groups ANCOVA (with prior knowledge as a covariate) indicated a significant interaction effect between PA and voice, $F(1, 29) = 4.42, p = .046$.

5 DISCUSSION, CONCLUSION, AND NEXT STEPS

This ongoing study investigates the effects of PA type (AI-generated vs. human) and voice (AI-generated vs. human) on learners' agent perception, attention, retention, and cognitive load. Preliminary findings from pilot study 1 indicated that retention and cognitive load were influenced by both the PA appearance, but not attention. In terms of agent perceptions, the results indicated that the type of PA influenced learners' perception of the agent's ability to facilitate learning, its human-like qualities, and its engagement level. However, it did not affect its credibility. This suggests that specific attributes of AI-generated PAs influence learners' perception of the PA in its ability to facilitate learning, human likeness, and engagement. While no significant effects of PA type and voice were found to contribute to attention, the type of voice was found to significantly impact cognitive load and retention. This suggests that voice is a factor to consider when designing and generating educational videos with AI. Furthermore, the results of this study suggest that English proficiency and prior knowledge are important factors to consider when considering using AI-generated PAs in instructional videos. Further research is essential to explore additional factors and contexts that may influence the effectiveness of AI-generated instructional agents and their applications in education.

Expanding upon preliminary findings and incorporating revisions to the research design of the above-mentioned pilot study 1, a modified full study with a larger sample size will be conducted as a next step. The target sample size for the full study is 140 participants, which is 35 participants per condition (4 conditions in total). Following the full study, future research could expand beyond this research and delve deeper with more diverse populations (i.e. K-12, special education, English as a foreign language), and various learning contexts (i.e. informal learning, different types of knowledge). In the long term, broadening this research to encompass more diverse populations, including K-12, special education, and English as a foreign language would be transformational in educational practice, as well as making theoretical contributions to previous research on PAs. Moreover, extending the study across various learning contexts will enhance the generalizability of the findings.

to other domains, which is critical at this point in time, especially with more GenAI video tools becoming accessible. Findings will inform and drive future design and development directions of AI and HCI practices, and ultimately contribute to narrowing the gap between human-computer interaction research and education.

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